18.4 Beyond denial: guiding agent behavior with policy constraints

When ACME first implemented the policy-aware loop described in the previous section, the system behaved much like a traditional authorization system. Each time the agent attempted to invoke a tool, the policy engine evaluated the request and either permitted or denied it. This approach was correct, but it revealed a practical limitation.

Agents do not follow a single path to a solution. They explore multiple strategies, trying different actions as they work toward a goal. When an action is denied, the agent revises its plan and tries something else. In some cases, the agent might repeatedly attempt actions that the policy would never allow.

From the policy system's perspective, everything was working as intended. From the agent's perspective, however, the system provided very little guidance. The agent learned what not to do by trial and error.

ACME’s engineers realized that denial alone was not enough. If policy was going to govern agent behavior effectively, it needed to do more than block unsafe actions. It needed to help the agent understand which actions were possible before it attempted them.

> ***NOTE***
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> Appendix C.4 provides a working demonstration of this pattern. The constraint-aware planning demo shows how partial requests produce policy residuals that the agent can use to guide its planning before executing actions.

18.4.1 From reactive enforcement to constraint-aware planning

To address this, ACME extended the role of the policy system. Instead of only using it to evaluate fully specified authorization requests, they enabled